C U Vivek

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Summary

Senior Performance Engineer with over 10 years of experience architecting Non-Functional Testing (NFT) strategies, distributed system scalability, and enterprise resiliency across Banking, Telecom, and Energy domains. Track record scaling American Express Payment Network Authorization systems to 6,000+ TPS, architecting automated chaos & resiliency validation with Toxiproxy, and engineering autonomous AI performance agents using LangGraph & RAG. Recipient of the American Express Centurion Award 2025.

Experience

Senior Performance Engineer - Senior Engineer II, American Express – Bengaluru, Karnataka, India — Feb 2025 – present

- Architected and deployed an internal AI Performance Agent using **LangGraph**, Vector Databases, and Ollama/Open-Source LLMs to automate performance test execution, multi-dimensional telemetry analysis, and conversational RAG diagnostics of historical defect patterns.
- Spearheaded enterprise scalability and fault-tolerance initiatives for high-throughput payment authorization services.

Senior Performance Engineer - Senior Engineer I, American Express – Bengaluru, Karnataka, India — Mar 2023 – Feb 2025

- Awarded the **Centurion Award 2025**, American Express's premier internal engineering recognition, for technical excellence and impact in system resiliency.
- Led performance engineering for American Express Payment Network Authorization systems, achieving horizontal scalability to **6,000 TPS** via microservices tuning, workload modeling, and thread/heap dump profiling.
- Containerized **Toxiproxy** to simulate network degradation, packet jitter, and latency injection; designed automated chaos engineering experiments across distributed microservices to eliminate resiliency gaps.
- Architected a Shift-Left Performance Testing framework integrated with **GitHub Actions**, enforcing automated throughput and latency validation gates during CI/CD to eliminate post-production defects.

Performance Engineer - Engineer III, American Express – Bengaluru, Karnataka, India — June 2021 – Mar 2023

- Executed large-scale distributed load tests using Apache JMeter and Grafana k6 across cloud-native container platforms.
- Monitored and diagnosed distributed transaction traces, connection pools, and JVM garbage collection behavior using Dynatrace and JProfiler.

Performance Engineer (Amdocs Core R&D Division), Amdocs India LLC – Pune, Maharashtra, India — Nov 2019 – June 2021

- Certified monthly drop releases for microservices-based digital applications on **Red Hat OpenShift (Kubernetes)** and **AWS EKS** environments.
- Executed end-to-end performance test monitoring, profiling, and bottleneck analysis across Couchbase, Apache Kafka, Prometheus, Grafana, and AWS services (EC2, RDS, MSK, CloudWatch).
- Conducted Load, Stress, Stability, Scalability, High Availability (HA) Failover, and Chaos Monkey resiliency experiments.
- Analyzed JVM thread and heap dumps with APM tools including Dynatrace, JProfiler, and JVisualVM to size production systems with enterprise architects.

Performance Engineer, Petrolink Data Services Pvt. Ltd. – Pune, Maharashtra, India July 2019 – Oct 2019

- Formulated Non-Functional Requirements (NFRs) and authored comprehensive Performance Test Plans.
- Engineered end-to-end user traversal test suites using Apache JMeter and profiled server CPU and memory footprints.

Application Development Analyst, Accenture Solutions Pvt. Ltd. – Pune, Maharashtra, India Dec 2016 – July 2019

- Developed performance test scripts using HP / Micro Focus LoadRunner across Web (HTTP/HTML) and Web Services protocols.
- Administered StormRunner cloud testing platform, configuring load generator provisioning and debugging execution anomalies.
- Conducted extensive load, stress, and endurance scenarios, analyzing server resource utilization and response times in Dynatrace.

Honors and Awards

Centurion Award 2025 2025

- Highest annual engineering recognition at American Express awarded for outstanding technical leadership, resilience engineering, and payment authorization scalability.

Certifications

AI Agents with LangGraph

Hands-On AI: Build a RAG Model from Scratch with Open Source

Microsoft Certified: Azure Fundamentals

AWS Certified Solutions Architect – Associate (SAA)

AWS Certified Cloud Practitioner (CLF)

Education

Vellore Institute of Technology (VIT University), Bachelor of Engineering (B.E.) in Civil Engineering – Vellore, Tamil Nadu, India July 2012 – June 2016

- Graduated with CGPA: 8.86 / 10

Skills

AI & Autonomous Agents: LangGraph, RAG (Retrieval-Augmented Generation), Vector Databases, Ollama, Open-Source LLMs, Autonomous Telemetry Diagnostics

Performance & Load Testing: Apache JMeter, Grafana k6, Micro Focus LoadRunner, StormRunner, Workload Modeling, Capacity Planning

Observability & APM: Dynatrace (OneAgent), Grafana, Prometheus, JProfiler, JVisualVM, JConsole, Heap & Thread Dump Analysis, GC Tuning

Resiliency & Chaos: Toxiproxy, Chaos Engineering, Chaos Monkey, Latency & Jitter Injection, Failover & High Availability (HA)

Cloud & Containers: Kubernetes, Red Hat OpenShift, AWS (EKS, EC2, RDS, MSK, CloudWatch), Docker

Distributed Data & CI/CD: Apache Kafka, Couchbase, Python, Bash Shell Scripting, GitHub Actions, Jenkins